

Java Collections - Exercises

* Create Class Employee with specific fields(id, name, salary, age).

List Operations

1. Create an **ArrayList<Employee>** and perform the following:

- Add 5 employees to the list.
- Remove an employee by index.
- Sort employees by salary.
- Find an employee with the highest salary.
- Convert the list to a **LinkedList<Employee>**.

2. Implement a program to:

- Insert 10 employee names in a **LinkedList**.
- Retrieve the 3rd employee in the list.
- Replace an employee at index 2.
- Remove an employee by name.

Set Operations

1. Create a **HashSet<Employee>** and do the following:

- Add 5 employees with different IDs.
- Try adding a duplicate employee (same ID & name).
- Print all employees and check if duplicates exist.
- Check if a specific employee exists in the set.

Map Operations

1. Create a `HashMap<Integer, Employee>` and do the following:

- Add 5 employees using ID as the key.
- Retrieve an employee by ID.
- Remove an employee using their ID.
- Print all employee details.

Stack Operations (`Stack<Employee>`)

- ◆ Use a `Stack<Employee>` to perform **LIFO operations**.

Push **5 employees** onto the stack.

Pop an employee from the stack (remove the last added employee).

Peek at the **top employee** without removing it.

Search for an employee in the stack.

Check if the stack is **empty**.

**** Explore more methods in the Collection framework.****